Airplane Crashes Analysis & Insights Platform

Graduation Project – Project Planning & Management Documentation

Team: [2] | Supervisor: [Dr.amaar mostafa] | Academic Year: [2026]

Project Planning & Management

1. Project Proposal

1.1 Overview

Air travel is among the safest forms of transport, yet every accident carries a high human and economic cost. Data on airplane crashes exists across many sources, but it is scattered, inconsistent and hard to learn from. This project collects, cleans and analyzes historical airplane crash data to uncover patterns in causes, timing, locations and aircraft types, and presents the findings in an interactive dashboard.

1.2 Problem Statement

There is no single, clean and easy-to-explore view of airplane crash data that answers questions such as:

- What are the most frequent causes of crashes?
- Is aviation safety improving over the decades?
- Which regions, operators and aircraft types are most affected?
- Which factors (weather, human error, technical failure) contribute most?

1.3 Objectives

- Collect and consolidate crash data from reliable public sources.
- Clean and standardize the data (missing values, duplicates, inconsistent categories).
- Perform exploratory data analysis (EDA) across time, geography, cause and aircraft type.
- Build an interactive dashboard that presents key insights.
- Build a machine learning model to classify or predict crash causes or severity.
- Deliver a final report with findings and recommendations.

1.4 Scope

scope:

- Historical commercial and passenger aviation crashes .
- Data cleaning, analysis, visualization and a final report.
- Interactive dashboard (Power BI /tableau/looker).

1.5 Stakeholders and Beneficiaries

Airlines, aviation safety authorities, researchers, students and aviation enthusiasts.

1.6 Deliverables

- Cleaned and documented dataset
- EDA notebooks and analysis report
- Interactive dashboard
- ML model
- Final documentation and presentation

1.7 Data Sources

- Aviation Safety Network (Scraping)
- Kaggle : Airplane Crashes Since 1919
- NTSB database

2. Project Plan

2.1 Timeline (Gantt Overview)

The plan follows the official submission dates, from 16 Oct 2026 to the final presentation on 4 Dec 2026.

Phase	Activity	16-22 Oct	23-29 Oct	30 Oct-6 Nov	7-13 Nov	14-20 Nov	21-30 Nov	1-4 Dec
1	Project planning and management	[redacted]						
2	Literature review	[redacted]						
3	Requirements gathering	[redacted]						
4	System analysis and design		[redacted]	[redacted]				

5	Data collection and cleaning		[redacted]	[redacted]	[redacted]			
6	EDA and dashboard implementation				[redacted]	[redacted]	[redacted]	
Phase	Activity	16-22 Oct	23-29 Oct	30 Oct-6 Nov	7-13 Nov	14-20 Nov	21-30 Nov	1-4 Dec
7	ML model					[redacted]	[redacted]	
8	Testing						[redacted]	[redacted]
9	Final report and presentation						[redacted]	[redacted]

2.2 Milestones

#	Milestone	Deadline
M1	Project planning and management, literature review and requirements gathering submitted	16 Oct 2026
M2	System analysis and design submitted	6 Nov 2026
M3	Implementation (source code and execution) delivered	30 Nov 2026
M4	Final presentation, testing and reports delivered	4 Dec 2026

2.3 Deliverables per Phase

Phase	Deliverable

Planning	Project proposal, project plan, risk plan
Data collection	Raw dataset and source list
Data cleaning	Clean dataset and data dictionary
EDA	Analysis notebooks and insights summary
Development	Dashboard and (optional) ML model
Testing	Test report and fixes log
Closure	Final documentation and presentation

2.4 Resource Allocation

Resource	Details
Human	Team members [5] and academic supervisor
Software	Python (Pandas, Matplotlib, seaborn, numby), SQL, Power BI, GitHub, tableau, looker
Budget	Minimal, using free and open-source tools

3. Task Assignment and Roles

The whole team works on every task together. No section, phase or deliverable is owned by a single person: all members join each working session, share the workload equally across every phase, and review every deliverable before it is submitted.

Working agreements:

- Weekly team meeting and a short progress update to the supervisor.
- Tasks tracked on a shared board (Notion / GitHub Projects).
- All code and files version-controlled on GitHub.

4. Risk Assessment and Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
R1	Poor data quality (missing values, errors)	High	High	Cross-check sources, apply documented cleaning rules, state assumptions

R2	Inconsistent categories across sources	High	Medium	Build a standard mapping for causes, aircraft types and countries
R3	Biased or incomplete historical records	Medium	Medium	Document limitations and avoid overclaiming in conclusions
R4	Scope creep	Medium	High	Freeze scope after M1; treat ML as optional
#	Risk	Likelihood	Impact	Mitigation
R5	Delays or uneven workload in the team	Medium	High	Weekly follow-up, clear ownership, buffer time before deadlines
R6	Tool or licensing issues	Low	Medium	Have a fallback (Python dashboards, Streamlit, free tools)
R7	Data loss or file conflicts	Low	High	GitHub version control and regular backups
R8	Weak ML performance	Medium	Low	Keep ML; focus on analysis and dashboard value

5. KPIs (Key Performance Indicators)

Area	KPI	Target
Data quality	Data completeness (key fields filled)	≥ 95% after cleaning

Data quality	Duplicate records remaining	0
Schedule	Milestones delivered on time	≥ 90%
Schedule	Task completion rate per sprint/week	≥ 85%
Analysis	Number of validated insights answering the research questions	≥ [10]
Dashboard	Page load / response time	< 3 seconds
Dashboard	Uptime / availability	≥ 99%
Dashboard	User satisfaction from test users (survey)	≥ 4 / 5
Adoption	Number of test users who try the dashboard	≥ [20]
Area	KPI	Target
ML	Model accuracy / F1-score	≥ [80%]
Documentation	Sections completed and reviewed by supervisor	100%

Literature Review

1. Suggested Improvements

The team plans to strengthen the project in these areas:

- Add an interactive map showing crash locations.
- Add dashboard filters for year range, region, aircraft type, operator and phase of flight.
- Group crash causes into a small standard set (human error, technical failure, weather, sabotage, other).
- Normalize crash counts by flight volume where data is available, so trends are not misleading.
- Classify causes from the crash summary text using NLP.
- State data limitations and possible bias openly in the report.

The lecturer's suggested improvements are added to this list after review.

2. Submission Schedule

Item	Deadline
Project Planning & Management	16 Oct 2026
Literature Review	16 Oct 2026
Requirements Gathering	16 Oct 2026
System Analysis & Design	6 Nov 2026
Implementation (Source Code & Execution)	30 Nov 2026
Final Presentation, Testing & Reports	4 Dec 2026

Requirements Gathering

1. Stakeholder Analysis

Stakeholder	Role	Main needs	Influence
Airlines and operators	Use the findings	Recurring causes and risk patterns to	High
Stakeholder	Role	Main needs	Influence
		improve safety procedures	
Aviation safety authorities	Use the findings	Trends by region, period and aircraft type to guide regulation	High
Lecturer / supervisor	Evaluates the project	Complete documentation and a working system on time	High
Project team	Builds the project	Clear scope, reliable data and an achievable plan	High
Researchers and analysts	Use the data	Clean dataset, filters and export	Medium
Students and the public	Learn from the project	Simple, understandable visuals	Low

2. User Stories & Use Cases

User Stories

ID	As a...	I want to...	So that...
US-1	Safety analyst	See the most common crash causes	I can focus prevention on what happens most
US-2	Authority officer	Filter crashes by year range and region	I can compare safety across places and periods
US-3	Analyst	Compare aircraft types and operators	I can spot higher-risk groups
US-4	Researcher	Download the cleaned dataset	I can run my own analysis
US-5	Student	View simple summary charts	I can understand the main findings quickly
US-6	Analyst	See a predicted cause for a crash	I can test the model on new cases

Use Cases

ID	Use case	Actor	Main flow	Outcome
UC-1	View dashboard overview	Any user	Open the dashboard and read the summary KPIs and charts	User sees the overall picture
UC-2	Filter and explore	Analyst, officer	Select filters (year, region, aircraft type, cause) and the charts update	User sees results for the chosen group
UC-3	Export data	Researcher	Choose the filtered view and export it	User gets the data file

UC-4	Predict cause	Analyst	Enter crash details and run the model	User gets a predicted cause

3. Functional Requirements

ID	Requirement	Priority
FR-1	The system collects crash data from the selected public sources	Must
FR-2	The system cleans the data: duplicates, missing values and inconsistent categories	Must
FR-3	The system standardizes crash causes, aircraft types and countries	Must
FR-4	The dashboard shows crashes over time	Must
FR-5	The dashboard shows crashes by cause, region and aircraft type	Must
ID	Requirement	Priority
FR-6	The dashboard lets users filter by year, region, aircraft type and cause	Must
FR-7	The dashboard shows key figures: total crashes, fatalities, trend	Should
FR-8	Users can export the filtered data	Should
FR-9	The dashboard shows crash locations on a map	Could
FR-10	A model predicts the likely cause or severity of a crash	Could

4. Non-functional Requirements

Category	Requirement	Measure
Performance	Dashboard pages load and respond quickly	Under 3 seconds
Reliability	Dashboard stays available when hosted	99% uptime
Security	Only public, non-personal data is used; edit access limited to the team	No personal data stored
Usability	A new user understands the dashboard without training	4 out of 5 in user testing
Data quality	Key fields are filled after cleaning	At least 95% complete
Maintainability	Code and data steps are commented and kept on GitHub	All files version-controlled
Compatibility	Works on current versions of Chrome, Edge and Firefox	Tested on all three